DAY 02 | PRE-READ

Organising Things

A light read before Day 2. About 15 minutes, no code required.

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Why read this?

Yesterday you gave the computer instructions one line at a time. Today you will learn to group instructions, organise information, and save it so it is still there tomorrow. These are ideas you already use at home and at work.

1. Functions are like kitchen appliances

A blender has one job. You put fruit in, press a button, and get a smoothie out. You do not need to know how the motor works each time. A function is the same: you give it inputs, it does a job, and it hands back a result. Write it once, use it whenever you need it.

> **Key idea:** inputs go in, a result comes out. The steps in the middle are hidden inside a name.

2. Four ways to organise things

Real life	Python name	What makes it special
A shopping list	list	Ordered, you can add, remove or change items.
Your date of birth (day, month, year)	tuple	Fixed. Once written, it does not change.
Unique stamps in a collection	set	No duplicates allowed.
A phone contact (name, number, email)	dictionary	Find information by a label, called a key.

3. Real information is nested

Think of a school register. It is a list of students, and each student has a name, a roll number and marks. That is a list of records, where each record has labels. This shape, a list of labelled records, is how almost all real data looks, from bank statements to online orders.

4. Why save to files?

When a program ends, everything in memory disappears, just like writing on a whiteboard that gets wiped. Saving to a file is like writing it in a notebook. Tomorrow you will learn two very common file types:

- **CSV**: a simple table, like a spreadsheet saved as plain text. Opens in Excel.
- **JSON**: a neat way to store labelled, nested information. Almost every website and app uses it to send data.

5. Think about these before class

- What everyday tasks do you repeat that could be a single "button"?
- Write down the details you store for one contact on your phone. Which are labels and which are values?
- Which of your daily lists must stay in order, and which just need to avoid duplicates?

Seiri seiton (整理整頓): keeping things sorted and in order